

Image Inpainting via Jointing Structure Restoration and End-to-end Reversible Diffusion

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Abstract

For pixel-level inpainting of large-area missing regions, existing methods often face inferior global consistency and blurred details, hampering visual-semantic consistency between inpainted and unmasked regions. Diffusion-based inpainting schemes, although partially mitigating the aforesaid issues, still exhibit limitations including high training costs and low inference efficiency. To address these challenges, this paper proposes a novel method, Inpainting via Jointing Structure Restoration and End-to-end Reversible Diffusion, which pioneers the application of an end-to-end reversible diffusion architecture to the inpainting task. For the inpainting task, the proposed method redesigns the sampling steps and reframes the optimization objective of diffusion models from noise prediction to direct image reconstruction; simultaneously, it introduces a discriminator to effectively alleviate the over-smoothing issue caused by the adjustment of this optimization objective. The Channel Attention Dynamic Injector aims for multi-level dynamic fusion of texture and edge structure priors into diffusion network layers, alleviating the separation between structural priors and generation process. Reversible modules are embedded within the diffusion network and between sampling steps to dynamically reconstruct intermediate features, thus reducing the training memory requirement to the level of a single consumer-grade GPU. Furthermore, through the joint

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training of the edge restoration network and diffusion inpainting network, the consistency between global texture and edges is ensured. Experimental results demonstrate that for large-area inpainting on the CelebA-HQ, Inp-JSRDiff achieves 0.2 dB higher PSNR and 12% lower FID than SOTA models, with efficient validation on datasets Places2 and Paris StreetView. Code is available at <https://github.com/bald-dog/Inp-JSRDiff.git>

Keywords: Image Inpainting, Reversible Diffusion, Structure and Texture, End-to-End Networks

1. Introduction

Image inpainting aims to reconstruct pixels in masked regions and ensure that predicted pixels in these regions maintain visual and semantic consistency with content in unmasked regions [1–3]. Inpainting techniques have been extensively applied in object removal, video restoration, cultural relic restoration [4], and image editing [5]. Existing image inpainting methods can be categorized into training-free inpainting methods and training-based deep learning inpainting methods. Training-free inpainting methods utilize content in unmasked regions within a single image to reconstruct masked regions. Such methods are classified into a) diffusion-based methods [6, 7] and b) patch-based methods [8, 9]. However, in cases of large-area missing content, limited by the sparsity of known information in a single image, these methods fail to handle irregular or complicated masks. Thus, inpainting methods have shifted toward data-driven deep learning training. Through the improvement of convolution strategies [10–13], efficient learning of general features for a category of images is enabled, allowing such methods to perform inpainting for this category. However, when large-area content is missing, constrained by the representational capabilities of convolutional networks, inpainted outcomes tend to exhibit poor semantic consistency and blurred details. To address these challenges, many studies have begun exploring solutions from two perspectives as follows.

a) Providing structural priors for image inpainting. By incorporating auxiliary signals including contours, sketches [15], or semantic features [16] into the inpaint-

ing network, geometric or topological constraints are provided for occluded regions. However, such methods [14, 19] have limitations. Most structure-guided inpainted are optimized in a stage-wise manner, where the structure predictor is supervised by proxy edge/contour objectives, while the final reconstruction/perceptual objectives are mainly imposed on the texture stage; this weak coupling makes it difficult for reconstruction-driven gradients to correct structural errors, especially under large masks. Moreover, structural cues are often injected by concatenation or shallow fusion, and their influence can be progressively weakened in deep semantic layers, making it hard to maintain global texture–edge consistency.

b) Using generative artificial intelligence for image inpainting. Ho et al. [17] proposed DDPM, which iteratively denoises Gaussian noise to approximate the data distribution, and this paradigm has been extended to inpainting for synthesizing missing content. Song et al. [33] further proposed DDIM for faster sampling, and Rombach et al. [27] introduced latent diffusion (LDM) to improve efficiency by denoising in a learned latent space. These diffusion models provide strong generative priors and have shown notable progress in semantic generation for inpainting [18–20]. Notably, large-hole inpainting is inherently multi-modal; diffusion can be viewed as iterative refinement that is often more controllable for global semantic coherence than one-shot adversarial regression. Nevertheless, extensive sampling operations still lead to high training and inference costs, making cost reduction and efficiency improvement key research directions.

Recently, researchers have begun exploring structure-guided diffusion inpainting models that explicitly incorporate structural cues (e.g., edges) to better preserve geometry and mitigate semantic inconsistency. For example, StrDiffusion [19] reformulates the denoising objective under structural guidance and adopts adaptive resampling to tackle the semantic discrepancy between masked and unmasked regions. Despite improved structural plausibility, such methods still inherit the multi-step sampling nature of diffusion and thus remain computationally expensive.

In contrast, we target few-step, end-to-end trainable inpainting with strict known-pixel consistency and practical memory efficiency. To address weak structure-texture coupling and the high cost of diffusion-based inpainting, we propose Inp-JSRDiff,

which integrates an end-to-end reversible diffusion framework [21] with joint structure restoration. We treat deterministic sampling as an unrolled reconstruction network with hard known-pixel clamping and learnable step coefficients $\{\alpha_t\}$, and reformulate the diffusion objective from noise prediction to direct masked image reconstruction, enabling few- or single-step inpainting; an adversarial discriminator is introduced to alleviate over-smoothing. To enforce global texture-edge consistency, we propose CADI to inject edge/texture priors across diffusion layers and jointly train an edge restoration network with the diffusion backbone to strengthen gradient coupling. Finally, reversible modules reconstruct intermediate features on-the-fly, reducing training memory to a single consumer-grade GPU. The contributions are summarized as follows:

1. For the first time, we introduce the end-to-end reversible diffusion framework into the image inpainting task and propose Inpainting via Jointing Structure Restoration and End-to-end Reversible Diffusion (Inp-JSRDiff). By reconstructing the sampling steps and shifting the diffusion model’s optimization objective from noise prediction to direct image reconstruction to better adapt to pixel-level tasks.
2. The Channel Attention Dynamic Injector (CADI) is designed to enable dynamic weight regulation and deep fusion of texture priors and edge structural priors into multiple levels of the diffusion network. Reversible modules are embedded within the diffusion network and between sampling steps to dynamically reconstruct intermediate features, thus reducing the training memory requirement to the level of a single consumer-grade GPU.
3. A task-adaptive multi-dimensional collaborative training strategy is proposed. Cross-module collaborative optimization is performed, a discriminator is introduced to mitigate the over-smoothing caused by the change of the optimization objective. An edge recovery network is constructed, which shares gradients and is jointly trained with the diffusion inpainting network.
4. Efficient inpainting with diffusion models under few-step sampling is achieved. The proposed framework has been validated on three major public datasets: CelebA-HQ [22], Places2 [23], and Paris StreetView [24]. The inpainting re-

sults balance texture fineness, edge integrity, and global consistency, offering a novel solution for diffusion models to tackle the task of image inpainting.

2. Related Work

To effectively deal with large-area occlusions and improve inpainting quality in different scenarios, researchers have explored various mainstream methods for this problem, including methods based on convolution, methods based on structure guidance, and methods based on diffusion models. In this paper, we focus on large and irregular masks, where maintaining global structure–texture consistency and reducing computational cost are both critical.

2.1. Image Inpainting Method Based on Convolution Strategy

Convolution-based inpainting methods guide content generation for occluded regions by aggregating contextual features [25, 26, 31], enabling reconstruction of large-scale missing areas. In recent years, Liu et al. [11] proposed a partial convolution-based inpainting method. By adapting to hole morphology, it performs convolution only on valid pixels and dynamically updates masks to expand valid regions, thereby enabling effective inpainting of irregular large-area holes. Subsequently, Yu et al. [12] proposed a gated convolution-based free-form image inpainting method. By integrating the gating mechanism with masks to dynamically adjust feature weights, they enhanced the inpainting performance of free-form large-area holes. However, such methods focus on local region filling rather than understanding image context, which can easily lead to local plausibility but global semantic inconsistencies, when restoring large-scale missing areas. Even with enhanced feature reasoning mechanisms such as recurrent feature propagation and iterative context inference [40], long-range semantic coherence under large irregular masks remains challenging without explicit structural guidance. To address this, LaMa [36] utilizes Fourier convolution [13] to enhance global pattern capture; by combining local spatial convolution with global Fourier-domain convolution, it enables efficient modeling of long-range feature dependencies, yet sacrifices the preservation of local details. AOT [30] aggregates contextual information from different receptive fields through multi-scale convolutions, enhancing contextual reasoning capabilities while preserving local details. However, sparse inputs weaken long-range

feature correlations. Quan et al. [32] proposed a three-stage inpainting architecture featuring a coarse inpainting network with a large receptive field and local and global refinement networks with varying receptive field sizes. Related non-learning-based approaches further exploit nonlocal self-similarity and low-rank priors to propagate global structures [41], yet their reliance on handcrafted assumptions and iterative optimization limits scalability and adaptability to complex natural scenes. By constructing local and global inpainting models, this architecture improves the performance of image inpainting.

Despite the innovations in convolutional strategies, the aforementioned works have addressed multiple key issues in inpainting and attempt to tackle the difficulty of large-scale missing regions. However, under large and irregular masks, sparse valid pixels weaken long-range dependency modeling and global semantic correlation, which often leads to structural distortion and inconsistent global layouts. Without explicit structural guidance, convolution-based methods may fail to meet high-fidelity inpainting requirements.

2.2. *Structure-guided Image Inpainting Method*

To mitigate structural distortion in convolutional methods, researchers constrained the semantic consistency of texture generation by incorporating structural information as control signals [31]. Nazeri et al. [14] designed a two-stage generative adversarial network framework: First, the edge generator predicts the edge structure of the missing region. Then, using this edge map as a guide, it is fed into the image inpainting network along with the occluded image. The image inpainting network fills in the missing content, thereby enhancing the structural coherence of the restored image. Guo et al. [28] proposed a texture-structure dual-generation architecture. Through bidirectional feature interaction and contextual aggregation, the architecture realizes structure-guided texture synthesis and texture-aided structure reconstruction, thus improving the global consistency of image inpainting. Dong et al. [29] proposed an image inpainting method enhanced by an incremental Transformer architecture. By introducing mask position encoding and processing missing regions in stages to strengthen the guidance of positional information on global structure, the method improves the structural coherence

of the restored results. Wang et al. [16] proposed a multi-scale adaptive prior image inpainting method that enhances performance by adjusting prior information at each scale to adapt to corresponding feature restoration.

Although these approaches attempt to guide image inpainting using edge structural information to mitigate the disconnect between texture and structure, they still exhibit significant limitations in terms of feature coordination mechanisms and the effectiveness of edge constraints. First, many methods decouple structure estimation and texture synthesis into separate stages: the structure branch is mainly optimized with edge/contour supervision, while reconstruction/perceptual objectives are largely imposed on the final inpainting output, which weakens end-to-end gradient interaction under large masks. Second, strategies that incorporate edge priors through concatenation or shallow feature fusion are limited to supplementing the shallow feature space of the network, and their influence can be progressively weakened in deep semantic layers, making it difficult to ensure global consistency between texture and edge deep features.

2.3. *Image Inpainting Method Based on Diffusion Model*

Diffusion models, as a generative paradigm for image, have begun to be applied to inpainting tasks, yet still exhibit the trade-off between performance and cost. Lugmayr et al. [18] were the first to apply DDPM to image inpainting, proposing Repaint. This method dynamically adjusts the resampling strategy of masks to perform denoising updates only on known regions, thereby effectively enhancing the quality of the inpainted image. However, its inference process requires the cumulative execution of 4,570 step iterations, with a single inference taking up to 360 seconds on consumer-grade GPUs; this severe efficiency bottleneck restricts its practical deployment on platforms. To address the issue of semantic inconsistency between structure and texture in diffusion-based inpainting, Liu et al. [19] proposed a structure-guided diffusion inpainting method, which uses edge structures as auxiliary information to guide the texture denoising process and regulate the semantic correlations between structure and texture. However, this method adopts a two-stage strategy, using the same diffusion network to inpaint structure and texture separately. This not only results in disjointed

gradient updates between structure and texture and insufficient semantic synergy, but also significantly increases computational costs and training cycles. Corneau et al. [20] optimized computational efficiency from a dimensionality reduction perspective by proposing the LatentPaint method based on latent space diffusion models. This approach reduces computational complexity in the diffusion process through forward-backward fusion and explicit propagation within the pre-trained model’s latent space. However, the experiment relies on the complete StableDiffusion architecture. Transitioning from general image generation to inpainting requires fine-tuning and updating a large number of model parameters, and still necessitates support for 24GB or more of GPU memory.

Overall, diffusion-based inpainting offers strong generative priors for large-hole completion, yet existing methods are often either slow due to many sampling steps [18] or costly to train/fine-tune because of heavy backbones and memory demands [20]. Moreover, stage-wise structure-guided diffusion still suffers from weak structure–texture coupling and increased cost [19], which motivates our end-to-end reversible diffusion inpainting framework with joint optimization and few-step, memory-efficient training (Section 3). To address the limitations of the aforementioned three approaches, this paper proposes an edge-guided end-to-end reversible diffusion image inpainting scheme. Its design focuses on enhancing inpainting quality, strengthening the synergy between structure and texture, improving inference efficiency, and reducing training costs. The details are provided in Section 3 below.

3. Methods

The core architecture of the proposed Inp-JSRDiff model is shown in Fig. 1. Figure 1(a) shows RevDiffInpaint, the reversible diffusion inpainting model described in Section 3.1.2 hereinbelow, which is employed to restore textures in occluded images. By inserting reversible structures (Section 3.1.3) to dynamically reconstruct intermediate features during backpropagation, the hardware requirements for training are reduced. Figure 1(b) E-StructRM, the edge structure restoration model (Section 3.3), is employed to restore the edge structure of the occluded image and inject structure guidance into RevDiffInpaint via CADL. In Fig. 1(c), the Channel Attention Dynamic

Injector (CADI, described in Section 3.2) is used to inject structural and texture information guidance into multiple layers of the U-Net within RevDiffInpaint, enabling deep-level guidance of the generation process by physical information.

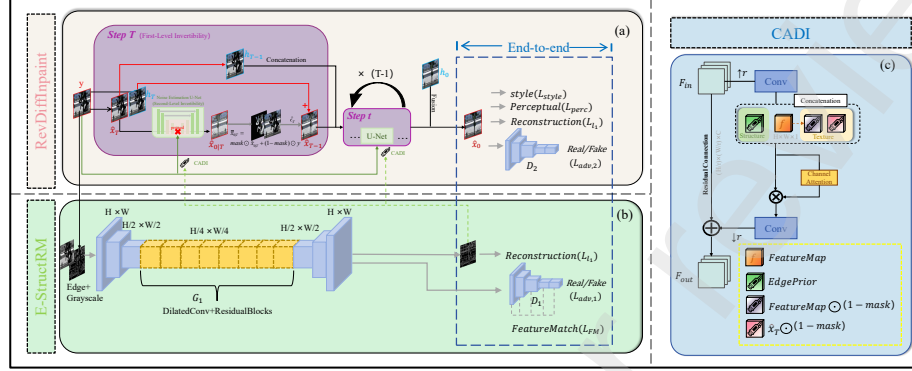


Fig. 1: Core architecture of the proposed Inp-JSRDiff model. (a) RevDiffInpaint, the reversible diffusion inpainting model (Section 3.1.2). (b) E-StructRM, the edge structure restoration model (Section 3.3). (c) Channel Attention Dynamic Injector, CADI (Section 3.2).

3.1. End-to-end reversible diffusion inpainting model

3.1.1. Principle of RevDiffInpaint Model

Forward diffusion and deterministic sampling. In variance-preserving diffusion models, the forward process admits the marginal form:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (1)$$

where $\bar{\alpha}_0 = 1$ and $\bar{\alpha}_t \in (0, 1]$. In our implementation, the per-step coefficients $\{\alpha_t\}_{t=1}^T$ are treated as learnable scalars, and

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s, \quad t = 1, \dots, T \quad (2)$$

This differs from conventional diffusion settings that prescribe $\{\alpha_t\}$ via a fixed schedule; here $\{\alpha_t\}$ are optimized jointly with the network parameters in an end-to-end unrolled reconstruction framework (Sec. 3.1.2).

Inpainting constraint. Given a binary inpainting mask $mask \in \{0, 1\}^{H \times W}$ (with $mask = 1$ indicating missing pixels), the observed image y provides valid pixel values on the known region $(1 - mask)$. We write this relation as:

$$y = (1 - mask) \odot x_0 \quad (3)$$

where $\odot$ denotes element-wise multiplication. The inpainting objective is to recon-

struct the missing region while keeping the known pixels consistent with y throughout the iterative process.

Reconstruction-oriented deterministic step (denoise $\rightarrow$ clamp $\rightarrow$ update). Starting from a measurement-conditioned initialization $\hat{x}_T = \phi(y)$ (i.e., the initial state is constructed from the observed image rather than pure Gaussian noise), the sampling procedure is unrolled from $t = T$ to $t = 1$. At each step t , we first predict the noise using a conditional network $\epsilon_\theta(\cdot)$:

$$\hat{\epsilon}_t = \epsilon_\theta(\hat{x}_t, t; \text{cond}) \quad (4)$$

Then we compute the denoised estimate:

$$\hat{x}_{0|t} = \frac{\hat{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\epsilon}_t}{\sqrt{\bar{\alpha}_t}} \quad (5)$$

To strictly enforce the inpainting constraint on known pixels, we apply an explicit pixel-clamping step on the denoised estimate:

$$\bar{x}_{0|t} = \text{mask} \odot \hat{x}_{0|t} + (1 - \text{mask}) \odot y \quad (6)$$

Property 1 (Exact consistency on known pixels). Eq. (6) guarantees $(1 - \text{mask}) \odot \bar{x}_{0|t} = (1 - \text{mask}) \odot y$ at every step.

Proof. Multiply both sides of Eq. (6) by $(1 - \text{mask})$ and use $(1 - \text{mask}) \odot \text{mask} = 0$.

Finally, we perform a deterministic DDIM-style update:

$$\hat{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \bar{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\epsilon}_t \quad (7)$$

Eqs. (4)-(7) define one unrolled reconstruction layer F_t , which maps $(\hat{x}_t, y, \text{mask}, \text{cond})$ to $\hat{x}_{t-1}$. The final output is $\hat{x}_0$ after iterating to $t = 0$.

3.1.2. End-to-end Inpainting Diffusion Model Framework

Unlike structure-guided diffusion inpainting methods (e.g., Repaint [18], StrDiffusion [19]), we adopt an end-to-end unrolled reconstruction framework and reinterpret the deterministic sampling procedure (Sec. 3.1.1) as an unrolled reconstruction network specialized for pixel-level inpainting. As illustrated in Fig. 1(a), the proposed update is repeatedly applied for T steps, resulting in a T -layer network $\mathcal{F} = F_1 \circ F_2 \circ \dots \circ F_T$. Each layer takes the current iterate $\hat{x}_t$, together with the observed image and the mask (and optional priors in “cond”), and outputs the next iterate $\hat{x}_{t-1}$:

$$\hat{x}_{t-1} = F_t(\hat{x}_t; y, \text{mask}, \text{cond}), \quad t = T, \dots, 1 \quad (8)$$

During training, the ground-truth image x_0 is available for supervision. We therefore directly optimize reconstruction-oriented objectives on the final inpainted output $\hat{x}_0$, while jointly fine-tuning the learnable diffusion coefficients $\{\alpha_t\}$ and the pre-trained weights in ϵ_Θ to adapt the entire inpainting process.

Loss functions. The loss of RevDiffInpaint consists of a pixel-level reconstruction term, an adversarial term to enhance high-frequency details, and optional perceptual/style terms to improve visual fidelity. Specifically, we employ the L_1 loss for pixel-level supervision:

$$L_1 = \frac{1}{N} \|\hat{x}_0 - x_0\|_1 \quad (9)$$

where N denotes the number of pixels.

To alleviate over-smoothing and encourage realistic textures, we introduce an adversarial loss $L_{\text{adv},2}$ with a discriminator $D_2(\cdot)$:

$$L_{\text{adv},2} = \mathbb{E}_{x_0} [\log D_2(x_0)] + \mathbb{E}_{\hat{x}_0} [\log (1 - D_2(\hat{x}_0))] \quad (10)$$

Meanwhile, we additionally introduce perceptual loss L_{perc} and style loss L_{style} :

$$L_{\text{perc}} = \mathbb{E} \left[\sum_i \frac{1}{N_i} \|\phi_i(\hat{x}_0) - \phi_i(x_0)\|_2 \right] \quad (11)$$

$$L_{\text{style}} = \mathbb{E} [\|G_j^\phi(\hat{x}_0) - G_j^\phi(x_0)\|] \quad (12)$$

Here, $\phi(\cdot)$ denotes the activation maps of a VGG-19 network pre-trained on ImageNet, $\phi_i(\cdot)$ is the activation of the i -th layer, and $G_j^\phi(\cdot)$ is the $C_j \times C_j$ Gram matrix constructed from the activation $\phi_j(\cdot)$.

Finally, the overall training objective of RevDiffInpaint is:

$$\mathcal{L}_{\text{RevDiffInpaint}} = \lambda_1 L_1 + \lambda_{\text{adv},2} L_{\text{adv},2} + \lambda_p L_{\text{perc}} + \lambda_s L_{\text{style}} \quad (13)$$

In our experiments, we set $\lambda_1 = 5.0$, $\lambda_{\text{adv},2} = 0.2$, $\lambda_p = 0.1$, and $\lambda_s = 0.1$.

3.1.3. Reversible Structural Design

To alleviate the memory bottleneck in end-to-end fine-tuning of diffusion models and reduce GPU resource requirements, we adopt two-order reversible design.

First-order reversibility between sampling steps [21]. As shown in Fig. 2, at sampling step t , the input $\hat{x}_t$ is directly transmitted to the next sampling block F_{t-1} through a bypass connection. Meanwhile, the output of the current block F_t is fused

with an auxiliary variable h_t (which carries intermediate states for reversible computation). Two learnable scaling coefficients u_t and v_t are applied to these two signals, respectively, with a constraint $u_t + v_t = 1$. Specifically, the forward mapping is defined as:

$$\begin{cases} \hat{x}_{t-1} = u_t F_t(\hat{x}_t; y, \text{mask}, \text{cond}) + v_t h_t, \\ h_{t-1} = \hat{x}_t \end{cases} \quad (14)$$

where $F_t(\cdot)$ denotes the unrolled reconstruction layer defined by Eqs. (4)-(7), and (y, mask) and “cond” are treated as available inputs rather than cached states. Given $(\hat{x}_{t-1}, h_{t-1})$, the original input pair $(\hat{x}_t, h_t)$ can be uniquely recovered by:

$$\begin{cases} \hat{x}_t = h_{t-1}, \\ h_t = [\hat{x}_{t-1} - u_t F_t(\hat{x}_t; y, \text{mask}, \text{cond})] / v_t \end{cases} \quad (15)$$

The bijective mapping in Eqs. (14)-(15) enables dynamic reconstruction of intermediate variables during backpropagation, thus reducing memory consumption with respect to the depth of unrolled sampling steps. To ensure dimensional consistency at the beginning and end of the network, learnable scalar parameters w_T and w_0 are introduced at the input and output ends, respectively:

$$h_T = w_T \hat{x}_T, \quad \hat{x}_0 = w_0 h_0 \quad (16)$$

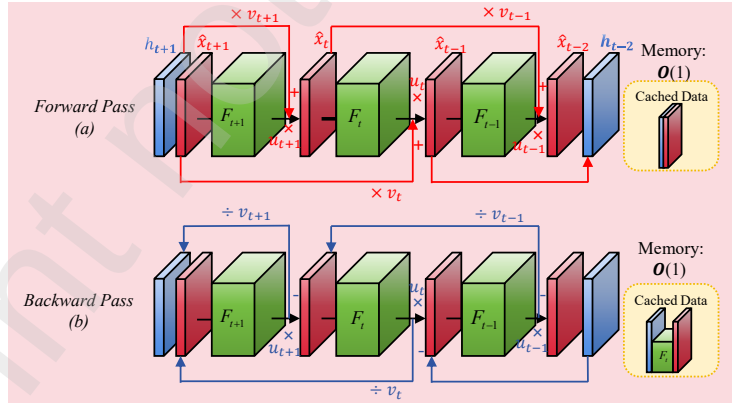


Fig. 2: Reversible structural design. (a) Forward process. (b) Backpropagation process.

In terms of memory management, during forward propagation we only cache the final output pair $(\hat{x}_0, h_0)$. During backpropagation, intermediate states are reconstructed layer-by-layer using Eqs. (14)-(15), and the reconstructed variables are released im-

mediately after computing the local gradients, so that the memory usage remains at the $O(1)$ level throughout the backward pass.

Second-order reversibility in the noise prediction network U-Net. The noise prediction U-Net contains multi-scale downsampling/upsampling blocks with skip and jump connections, which are typically memory intensive. To further enhance memory efficiency, as illustrated in Fig. 3(a), we extend the reversible connection technique to the block level of U-Net and implement grouping-and-concatenation for consecutive transformations within each block. Specifically, after trimming the SD-v1.5 backbone, the network contains 2 downsampling blocks (DB1-DB2) and 2 upsampling blocks (UB1-UB2). We adopt a structural grouping strategy aligned with the number and index positions of residual blocks and attention blocks within each block: DB1 and DB2 are divided into 2 groups, while UB1 and UB2 are divided into 3 groups. Within each group, we insert a CADI module after each residual/attention block, forming a reversible sequence “residual-block-CADI block-attention block-CADI block” as shown in Fig. 3(b). This second-order reversible design further reduces the memory footprint of the noise-prediction network during end-to-end training.

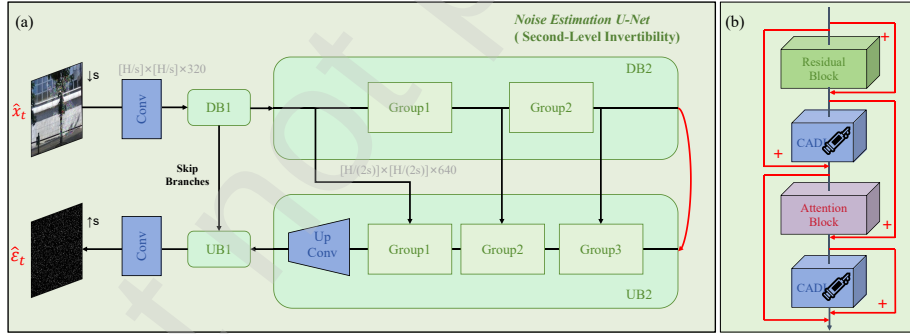


Fig. 3: U-Net network for noise estimation, based on the SD v1.5 model. (a) The U-Net noise prediction network after the SD v1.5 model is trimmed (the trimmed network contains 2 layers of downsampling and 2 layers of upsampling). The downsampling blocks DB1 and DB2, along with the upsampling blocks UB1 and UB2, all employ a structured grouping strategy. The grouping principle strictly aligns with the number and index positions of residual blocks and attention blocks within each block. Specifically, DB1 and DB2 are each divided into 2 groups, while UB1 and UB2 are each divided into 3 groups. (b) Within each group, add a CADI block after each residual block and attention block. Each group contains residual blocks (incorporated during channel matching), attention blocks, and CADI blocks corresponding to the respective indices, forming a sequence of "residual block-CADI block-attention block-CADI block".

3.2. Structural and Consistency Priors Dynamic Injector Based On Channel Attention

The dynamic injector design is illustrated in Fig. 1(c). By employing lightweight convolutions to bridge the feature domain and the image-aligned domain, we inject texture and structural priors into the intermediate features, enabling deep integration with high-level representations. In contrast to general inverse-problem formulations, our inpainting setting provides an explicit known-pixel consistency constraint: the observed image y serves as a fixed anchor on the known region $(1 - \text{mask})$, while step-dependent guidance extracted from intermediate features provides a dynamic consistency signal during denoising.

Texture anchor and known-pixel consistency guidance. Specifically, the observed image y (known pixels) serves as a static anchor reference, while the current-step reconstruction feature on the known region, $f \odot (1 - \text{mask})$, acts as a dynamic constraint, forming a known-pixel consistency regularization in the feature fusion stage. Structural information guidance is provided by the repaired edges from the edge-repair network, which are injected as explicit structural priors to guide the diffusion denoising process. We first map the intermediate deep feature to an image-aligned spatial descriptor:

$$f = \text{Conv}_1((F_{\text{in}})_r) \in \mathbb{R}^{H \times W \times 1} \quad (N = H \times W) \quad (17)$$

and then fuse the priors through a second convolution:

$$F_{\text{out}} = F_{\text{in}} + (\text{Conv}_2([f, f \odot (1 - \text{mask}), y, \text{edge_prior}]))_r \quad (18)$$

Among them, $F_{\text{in}} \in \mathbb{R}^{(H/r) \times (W/r) \times C}$ denotes the intermediate feature of U-Net, and $(\cdot)_r / (\cdot)^r$ denotes pixel rearrangement/inverse-rearrangement with a scaling ratio r . The concatenation $[f, f \odot (1 - \text{mask}), y, \text{edge_prior}]$ represents the channel-wise stacking of the spatial descriptor, its known-region response, the static texture anchor, and the structural prior (edge guidance).

On-channel attention for adaptive fusion. The channel attention mechanism enables dynamic adjustment of feature weights after fusion. Specifically, the fused feature F_{out} is first fed into global average pooling (GAP) to obtain a channel descriptor $F_{\text{gap}} = \text{GAP}(F_{\text{out}}) \in \mathbb{R}^{1 \times 1 \times C}$. Then a two-layer MLP with ReLU and Sigmoid is applied to generate channel weights. To avoid confusion with the learnable diffusion coefficients $\{\alpha_t\}$, we denote the channel-weight vector by $\mathbf{a} \in (0, 1)^C$:

$F_{\text{act}} = \text{ReLU}(W_1 \cdot \text{flatten}(F_{\text{gap}}) + b_1)$, $\mathbf{a} = \text{Sigmoid}(W_2 \cdot F_{\text{act}} + b_2)$, and the attention-weighted feature is obtained by channel-wise multiplication: $F_{\text{fusion}} = F_{\text{out}} \odot \mathbf{a}$. The proposed injector adopts a hierarchical injection strategy and is deployed after the residual modules and attention modules of each downsampling and upsampling layer in the U-Net, as shown in Fig. 3(b). This enables prior features to propagate effectively across multi-scale spaces, thereby achieving deep guidance for diffusion-based inpainting.

3.3. E-StructRM

In Fig. 1(b), E-StructRM is designed to restore edge structures, thereby providing explicit edge priors for RevDiffInpaint. E-StructRM consists of an edge repair network G_1 and a discriminator D_1 .

Architecture of G_1 . The edge repairer G_1 contains two downsampling encoders, followed by eight residual blocks, and a decoder that upscales the features back to the original resolution. Each residual block adopts a dual-convolution structure with reflection padding, a spectrally normalized 3×3 convolution, instance normalization, and a ReLU activation after the first convolution. An identity mapping is used to form residual connections.

Training data preparation and inputs. Let x_0 denote a real image. We extract its grayscale map I_{gray} and groundtruth edge map C_{gt} . Given the inpainting mask, the observed image is $y = (1 - \text{mask}) \odot x_0$. Accordingly, we extract the observed grayscale map and observed edge map from y , denoted as $\bar{I}_{\text{gray}}$ and $\bar{C}$, respectively. We feed $(\bar{I}_{\text{gray}}, \bar{C}, \text{mask})$ into G_1 to predict a complete edge map:

$$C_{\text{pred}} = G_1(\bar{I}_{\text{gray}}, \bar{C}, \text{mask}) \quad (19)$$

The edge-repair network is trained with an adversarial loss $L_{\text{adv},1}$ and a feature matching loss L_{FM} , forming the overall objective:

$$\mathcal{L}_{\text{edge}} = \min_{G_1} \max_{D_1} (\lambda_{\text{adv},1} L_{\text{adv},1} + \lambda_{\text{FM}} L_{\text{FM}}) \quad (20)$$

The adversarial loss is defined as:

$$L_{\text{adv},1} = \mathbb{E}_{(C_{\text{gt}}, \bar{I}_{\text{gray}})} [\log D_1(C_{\text{gt}} | \bar{I}_{\text{gray}})] + \mathbb{E}_{(\bar{I}_{\text{gray}})} [\log (1 - D_1(C_{\text{pred}} | \bar{I}_{\text{gray}}))] \quad (21)$$

and the feature matching loss is:

$$L_{\text{FM}} = \mathbb{E} \left[\sum_{i=1}^L \frac{1}{N_i} \left\| D_i^{(i)}(C_{\text{gt}}) - D_i^{(i)}(C_{\text{pred}}) \right\|_1 \right] \quad (22)$$

where $D_i^{(j)}(\cdot)$ denotes the activation of the i -th intermediate layer of D_1 , and N_i is the number of elements in that layer. In our experiments, $\lambda_{\text{adv},1}$ and λ_{FM} are the weight coefficients for the adversarial and feature matching terms, respectively.

3.4. Optimization Strategy

Our method first performs pre-training on the edge repair network for 5 epochs to mitigate the instability caused by the adversarial objective. Then, an end-to-end joint optimization strategy (as illustrated in Fig. 2(a) and Fig. 2(b)) is adopted to simultaneously optimize E-StructRM and RevDiffInpaint. To realize this end-to-end joint optimization, the overall loss function is formulated as:

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{RevDiffInpaint}} + \zeta \mathcal{L}_{\text{edge}} \quad (23)$$

where $\mathcal{L}_{\text{RevDiffInpaint}}$ is defined in Eq. (13), $\mathcal{L}_{\text{edge}}$ is defined in Eq. (20), and ζ is a balancing coefficient (set to 0.1 in our experiments).

Why joint optimization provides consistent gradients across modules. We jointly optimize E-StructRM and RevDiffInpaint by injecting the predicted edge prior C_{pred} into the unrolled diffusion process (Sec. 3.2) as a differentiable guidance signal. In this way, the final inpainting objective does not only supervise the reconstructed image $\hat{x}_0$, but also provides an explicit gradient path to update the edge repair network. Formally, letting ϕ denote the parameters of G_1 , the gradient of the overall objective w.r.t. ϕ follows the chain rule:

$$\frac{\partial \mathcal{L}_{\text{all}}}{\partial \phi} = \zeta \frac{\partial \mathcal{L}_{\text{edge}}}{\partial \phi} + \frac{\partial \mathcal{L}_{\text{RevDiffInpaint}}}{\partial \hat{x}_0} \cdot \frac{\partial \hat{x}_0}{\partial C_{\text{pred}}} \cdot \frac{\partial C_{\text{pred}}}{\partial \phi} \quad (24)$$

The second term indicates that the reconstruction loss backpropagates through the injected prior C_{pred} to G_1 , which yields consistent cross-module updates during end-to-end training.

4. Experiment

In this section, we first describe our experimental setup, dataset, research methodology, and implementation details. We then evaluate and analyze our approach through comparative experiments and ablation studies, demonstrating several practical applications. Finally, we examine the impact of the sampling step T on model mechanisms and identify limitations of current models.

4.1. Experimental Setup

4.1.1. Dataset

This paper conducts experiments on image inpainting tasks across three public datasets. Dataset CelebA-HQ [22] comprise 30,000 facial images. We randomly selected 28,000 images for training and reserved the remaining 2,000 for testing. Dataset Places2 [23] is a large-scale scene recognition dataset. We selected 20 categories (The specific settings are consistent with LGNet.) to construct the training dataset. Specifically, 2,000 images were randomly selected from the Places2 training set for each category, totaling 100,000 images in the training set. All images in the test set for these 20 categories (2,000 images in total) were used as our test set. Dataset Paris StreetView [24] consists of street-level images. Using the dataset’s original configuration, 14,900 images were employed as the training set and 100 images as the test set.

We employ NVIDIA’s irregular mask [11] as the mask for network training and testing. The irregular mask dataset comprises six categories with varying hole ratios: (0.01, 0.1], (0.1, 0.2], (0.2, 0.3], ..., (0.5, 0.6]. To ensure fairness, images for training and testing are randomly paired with masks of varying hole ratios.

4.1.2. Training Program

We implemented the proposed network architecture in PyTorch. This network is trained on images of size $3 \times 256 \times 256$ and masks of size $1 \times 256 \times 256$, respectively. All learnable parameters of the model are jointly trained in an end-to-end paradigm, which specifically include the shared structured U-Net₀ and its internal weighting factors (i.e., v_i , where $u_i \triangleq 1 - v_i$ in T sampling steps); non-shared and step-specific diffusion parameter sets $\{\alpha_i\}$; the family of weighting scalars $\{v_i\}$ in the reversible diffusion sampling pipeline (satisfying $u_i \triangleq 1 - v_i$); and scaling factors w_T and w_0 . The initialization values of the scalars α_i , v_i , v_T , w_T , and w_0 are set to 0.5, 0.5, 0.5, 1.0, and 0.0, respectively. All experiments are trained using the Adam optimizer. After preliminary experiments, we set the learning rate of the edge restoration network and the reversible diffusion inpainting network to $1e-4$, and the learning rate of the discriminator to $1e-5$. Our approach reuses the noise estimation U-Net pretrained on the LAION dataset from SD v1.5, further simplified by removing its temporal embedding,

cross-attention layers, and the last three scales to balance efficiency and performance.

In this paper, we compare our method with seven state-of-the-art baseline inpainting methods, as follows: AOT [30], LGNet [32], LaMa [36], MAPs [16], EdgeConnect [14], Repaint [18], and StrDiffusion [19]. We employed several common metrics from image inpainting tasks: l_1 , PSNR, SSIM [37], FID [38], and LPIPS [39]. The first three metrics are pixel-level indicators, while the latter two are perceptual-level metrics.

4.2. Comparison

4.2.1. Quantitative Comparisons

Table 1 presents quantitative comparisons between our method and several baseline methods on the Places2, CelebA-HQ, and PSV datasets using masks with varying porosities. It can be seen that our method demonstrates superior performance on both pixel-level metrics (L_1 , PSNR, and SSIM) and feature-level metrics (FID and LPIPS), indicating its advantages in both faithful reproduction and reasonable semantic generation. Notably, our method shows a more pronounced advantage under higher-porosity settings. Taking CelebA-HQ as an example, when the porosity lies in $[0.3, 0.6]$, our approach achieves an average $\sim 0.75\%$ improvement in the pixel-level fidelity metric PSNR over the best competing method, while reducing the perceptual quality metric FID by $\sim 9.5\%$ on average. These results indicate that, for larger missing regions, our method better balances pixel-wise reconstruction quality and overall perceptual coherence.

Furthermore, as shown in Table 2, compared with diffusion-based inpainting methods such as Repaint and StrDiffusion, our method requires fewer parameters (Among them, 115.2 M parameters come from the reversible diffusion structure of RevDiffInpaint, and 13.5 M parameters come from the generator of E-StructRM.). Although our method has a longer per-step inference latency (330 ms), which is higher than that of Repaint (78 ms) and StrDiffusion (68 ms), the use of very few sampling steps still gives our approach an advantage in overall inference speed. In Table 2, T denotes the number of sampling steps in our method after reconstructing the optimization objective and redefining DDIM sampling. We conducted experiments with T set to 1, 2, and 3.

Figure 4 shows the comparison of SSIM, PSNR, LPIPS, and FID metrics for our

Table 1: QUANTITATIVE COMPARISONS OF OUR METHOD WITH OTHER ADVANCED INPAINTING METHODS ON THREE DATASETS. $\uparrow$ HIGHER IS BETTER. $\downarrow$ LOWER IS BETTER.

Metrics	Method	Places2						CelebA-HQ						Paris StreetView					
		Mask ratio						Mask ratio						Mask ratio					
		0.01,0.1	0.1,0.2	0.2,0.3	0.3,0.4	0.4,0.5	0.5,0.6	0.01,0.1	0.1,0.2	0.2,0.3	0.3,0.4	0.4,0.5	0.5,0.6	0.01,0.1	0.1,0.2	0.2,0.3	0.3,0.4	0.4,0.5	0.5,0.6
SSIM $\uparrow$	AOT	0.965	0.911	0.850	0.788	0.709	0.608	0.985	0.957	0.915	0.865	0.827	0.682	0.977	0.960	0.920	0.860	0.806	0.694
	LQNet	0.969	0.916	0.856	0.789	0.717	0.636	0.995	0.981	0.964	0.942	0.916	0.839	0.987	0.970	0.930	0.878	0.830	0.708
	LaMa	0.968	0.919	0.857	0.792	0.723	0.619	0.981	0.960	0.913	0.893	0.856	0.790	0.978	0.958	0.918	0.864	0.813	0.700
	EdgeConnect	0.964	0.911	0.849	0.785	0.717	0.628	0.990	0.958	0.912	0.895	0.853	0.764	0.979	0.957	0.916	0.865	0.814	0.695
	MAPs	0.964	0.910	0.851	0.784	0.719	0.617	0.990	0.976	0.948	0.905	0.864	0.738	0.985	0.964	0.922	0.867	0.817	0.687
	Repaint	0.966	0.921	0.859	0.792	0.726	0.621	0.980	0.983	0.938	0.919	0.875	0.798	0.980	0.965	0.920	0.872	0.820	0.702
	StrDiffusion	0.967	0.918	0.856	0.788	0.723	0.624	0.992	0.970	0.946	0.939	0.890	0.829	0.983	0.963	0.928	0.879	0.822	0.698
	Inp-JSRDiff, $T=1$	0.973	0.920	0.862	0.795	0.729	0.634	0.993	0.984	0.967	0.941	0.907	0.829	0.989	0.968	0.935	0.891	0.826	0.709
	Inp-JSRDiff, $T=2$	-	-	-	-	-	-	0.995	0.986	0.969	0.948	0.913	0.835	0.990	0.971	0.938	0.893	0.832	0.712
	Inp-JSRDiff, $T=3$	-	-	-	-	-	-	0.996	0.988	0.971	0.950	0.917	0.842	0.992	0.972	0.938	0.892	0.831	0.712
PSNR $\uparrow$	AOT	32.58	27.29	24.43	22.36	20.78	18.11	34.41	30.98	25.63	23.22	21.30	18.66	34.06	30.97	27.88	25.35	23.58	21.31
	LQNet	33.49	27.84	24.57	22.31	20.53	18.51	40.04	32.89	29.85	27.63	25.74	21.99	37.10	32.38	28.65	26.06	24.26	21.88
	LaMa	33.53	27.96	24.51	22.44	20.62	18.19	34.69	31.75	25.78	24.73	22.64	19.13	35.92	31.06	27.95	25.86	23.51	21.45
	EdgeConnect	32.25	26.99	24.10	22.07	20.51	18.45	33.87	30.75	26.47	23.59	22.15	18.98	34.99	31.25	26.88	25.49	23.42	21.57
	MAPs	32.98	26.94	24.07	22.13	20.66	18.05	36.26	31.34	27.92	25.17	23.21	19.90	36.28	31.58	27.99	25.44	23.68	21.09
	Repaint	33.74	28.00	24.62	22.48	20.79	18.20	34.00	32.54	28.02	26.13	24.72	20.54	36.05	31.42	27.12	25.73	24.39	21.07
	StrDiffusion	33.75	27.95	24.41	22.39	20.73	18.42	38.28	33.04	29.13	26.89	24.97	21.34	37.11	31.86	27.94	25.98	24.73	21.68
	Inp-JSRDiff, $T=1$	34.56	28.32	24.97	23.04	20.81	18.49	40.98	34.46	30.65	27.83	25.58	22.04	38.06	32.24	29.01	26.27	24.43	21.90
	Inp-JSRDiff, $T=2$	-	-	-	-	-	-	41.02	34.79	30.84	27.89	25.69	22.14	38.12	32.35	29.35	26.38	24.86	21.97
	Inp-JSRDiff, $T=3$	-	-	-	-	-	-	41.18	34.83	31.07	27.93	25.82	22.19	38.15	32.41	29.87	26.34	24.90	21.95
LQO-2 $\downarrow$	AOT	0.53	0.95	1.53	2.97	3.06	5.32	0.52	1.15	2.03	3.03	4.21	6.32	0.54	0.95	1.57	2.63	3.93	6.22
	LQNet	0.47	0.68	1.18	1.61	3.23	4.86	0.46	0.73	1.20	1.71	2.33	3.52	0.52	1.02	1.58	2.72	3.65	5.91
	LaMa	0.52	1.07	2.00	2.26	3.12	4.73	0.54	1.17	2.12	3.11	4.28	6.73	0.53	1.01	1.65	2.74	3.82	6.19
	EdgeConnect	0.48	0.97	1.64	2.42	3.43	4.17	0.41	0.93	1.94	2.89	3.73	4.39	0.54	1.06	1.63	2.79	3.79	6.23
	MAPs	0.43	0.96	1.26	2.18	3.04	4.81	0.33	0.75	1.36	2.18	3.04	5.13	0.35	1.07	1.97	2.87	4.17	6.31
	Repaint	0.42	0.83	1.65	2.02	3.51	4.52	0.42	0.73	1.65	1.92	3.04	4.02	0.51	1.05	1.59	2.71	3.92	5.92
	StrDiffusion	0.40	0.70	1.23	1.83	2.98	4.48	0.39	0.69	1.19	1.83	2.88	3.78	0.59	1.03	1.55	2.74	3.87	5.89
	Inp-JSRDiff, $T=1$	0.20	0.54	1.16	1.77	2.57	3.98	0.20	0.56	1.08	1.76	2.57	3.78	0.51	1.05	1.82	2.73	3.79	5.71
	Inp-JSRDiff, $T=2$	-	-	-	-	-	-	0.20	0.53	1.02	1.72	2.37	3.59	0.48	1.02	1.64	2.64	3.62	5.68
	Inp-JSRDiff, $T=3$	-	-	-	-	-	-	0.19	0.49	0.99	1.69	2.29	3.46	0.47	0.99	1.55	2.59	3.58	5.62
LPIPS $\downarrow$	AOT	0.027	0.058	0.100	0.148	0.196	0.287	0.022	0.048	0.084	0.128	0.166	0.236	0.029	0.046	0.075	0.116	0.159	0.225
	LQNet	0.020	0.053	0.089	0.144	0.187	0.269	0.007	0.026	0.042	0.062	0.095	0.124	0.014	0.034	0.069	0.112	0.157	0.240
	LaMa	0.023	0.054	0.087	0.146	0.189	0.274	0.026	0.051	0.089	0.134	0.178	0.256	0.031	0.047	0.076	0.117	0.160	0.237
	EdgeConnect	0.022	0.056	0.088	0.143	0.186	0.284	0.023	0.043	0.057	0.121	0.153	0.224	0.035	0.044	0.079	0.116	0.164	0.241
	MAPs	0.024	0.057	0.089	0.146	0.188	0.289	0.015	0.034	0.066	0.105	0.145	0.229	0.018	0.039	0.076	0.120	0.163	0.249
	Repaint	0.021	0.052	0.088	0.142	0.186	0.283	0.018	0.029	0.058	0.093	0.109	0.143	0.028	0.041	0.072	0.115	0.159	0.237
	StrDiffusion	0.020	0.053	0.085	0.141	0.185	0.276	0.014	0.024	0.052	0.082	0.103	0.119	0.018	0.036	0.070	0.113	0.158	0.235
	Inp-JSRDiff, $T=1$	0.018	0.050	0.079	0.138	0.182	0.270	0.007	0.023	0.041	0.060	0.093	0.114	0.014	0.032	0.063	0.109	0.152	0.228
	Inp-JSRDiff, $T=2$	-	-	-	-	-	-	0.007	0.026	0.049	0.063	0.094	0.120	0.016	0.038	0.069	0.113	0.164	0.242
	Inp-JSRDiff, $T=3$	-	-	-	-	-	-	0.009	0.031	0.052	0.066	0.096	0.160	0.017	0.039	0.078	0.121	0.173	0.259
FID $\downarrow$	AOT	0.47	2.39	5.88	13.68	17.36	26.73	1.68	3.23	7.54	13.40	20.08	38.17	12.01	20.63	31.84	49.33	68.55	91.27
	LQNet	0.32	2.02	5.36	10.98	16.87	26.54	0.51	1.70	2.61	3.72	7.06	13.29	5.97	14.36	29.16	44.98	60.52	84.43
	LaMa	0.30	2.18	5.39	12.04	17.27	26.94	1.72	3.56	7.98	14.98	24.05	44.09	11.76	18.58	29.87	46.48	65.48	87.43
	EdgeConnect	0.41	2.23	5.68	13.32	17.70	26.86	1.64	3.09	8.54	12.09	18.54	24.52	12.08	17.39	39.18	47.17	66.35	89.12
	MAPs	0.43	2.26	5.67	13.02	17.47	26.94	0.97	2.22	4.24	7.37	10.79	18.90	6.93	16.49	31.27	48.42	66.87	87.51
	Repaint	0.42	1.95	5.28	10.78	16.62	25.82	0.82	2.88	3.97	7.04	9.76	17.03	8.72	17.24	28.42	45.53	59.94	82.74
	StrDiffusion	0.38	1.94	5.07	9.79	16.59	25.06	0.61	2.05	3.42	6.45	7.98	13.02	6.39	15.97	28.04	45.04	60.01	80.99
	Inp-JSRDiff, $T=1$	0.34	1.92	5.04	9.64	16.47	25.56	0.30	1.57	2.42	3.53	6.82	9.28	4.82	13.96	26.45	43.27	57.97	80.58
	Inp-JSRDiff, $T=2$	-	-	-	-	-	-	0.39	1.88	2.96	4.20	7.23	12.98	4.98	14.27	26.98	44.98	60.52	85.39
	Inp-JSRDiff, $T=3$	-	-	-	-	-	-	0.74	2.08	3.51	5.19	7.68	16.88	5.63	15.03	28.06	47.03	61.66	91.05

method with different T values and porosities in the CelebA-HQ dataset. It can be seen that our method demonstrates superior performance at the feature level when $T = 1$, while delivering better pixel-level restoration results at $T \geq 2$. The reasons are that we reconstructs the degree of freedom of DDIM through the number of sampling steps T . When $T = 1$, sampling completes the mapping from the noisy image $\hat{x}_T$ to the restored image $\hat{x}_0$ in one step, where β_t is set to a large value to rapidly attenuate noise, forcing the model to construct the global structure using the edge/texture prior injected by CADI. For $T \geq 2$, β_t adopts a linear decreasing schedule, allowing multi-step noise refinement, the pixel-level loss l_1 (Eq. (9)) contributes cumulatively with T

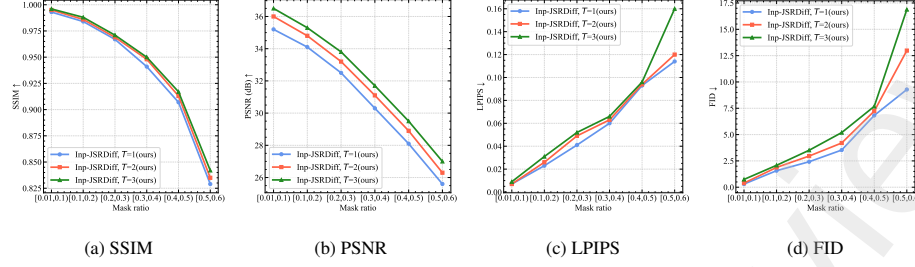


Fig. 4: Comparison of SSIM, PSNR, LPIPS, and FID metrics for our method with different T values and porosities in the CelebA-HQ dataset.

steps, shifting the model from prior-guided structure generation to data-driven pixel refinement. This coupled design of T -noise scheduling serves as the core for realizing the on-demand switching between perceptual quality and pixel accuracy.

Table 2: COMPARING OUR APPROACH WITH OTHER MODELS IN TERMS OF PARAMETERS AND INFERENCE TIME.

Method	AOT	LGNet	LaMa	EdgeConnect	MAPs	Repaint	StrDiffusion	Ours
Param(M)	15.2	115.0	50.98	22.0	47.30	527.2	384	128.7
Time(ms)	22	24	32	27	38	$73 \times 4570^{\text{iters}}$	$68 \times 100^{\text{iters}}$	$330 \times T$

4.2.2. Qualitative evaluation

Figure 5 presents the visual comparison results of 8 image inpainting methods on the PSV, CelebA-HQ, and Places2 datasets. Existing methods exhibit limitations to varying degrees: LAMA lacks global semantic understanding of damaged images, leading to insufficient structural consistency and blurred details in inpainting results. For instance, in the restoration of the character’s eye in Fig. 5(d), LAMA fails to accurately restore the shape of the eye socket and mistakenly generates a cyan eye region (which should actually be black). Although the contextual reasoning capability of AOT enhances the inpainting effect, it has deficiencies in natural color transition. In the inpainting of the grassland region in Fig. 5(h), both its texture details and color distribution show obvious differences from the original region, with significant inpainting traces. EdgeConnect and MAPs introduce single-scale and multi-scale priors to assist inpainting; however, due to the insufficient expressive capability of their image texture inpainting networks, the effectiveness of structural guidance is limited. For instance, in the inpainting of the ascending shark’s head in Fig. 5(g), neither method can accu-

rately depict this structure. The optimization framework of LGNet still retains visible artifacts, demonstrated in Fig. 5(f). RePaint relies on a probabilistic generation mechanism and tends to lack high-frequency details, resulting in over-smoothed inpainting results. For example, the wire mesh details in the inpainting, illustrated in Fig. 5(b), fail to be fully restored. StrDiffusion can achieve relatively good inpainting performance, while it has excessively high computational costs in training and inference, which is not conducive to practical deployment.

In summary, through the above comparison, our method can complete training and inference under controllable computational costs. CADI is used to dynamically inject prior to the RevDiffInpaint at multiple levels. As a result, the generation network not only possesses strong texture expression capability but also makes full use of structural guidance, thereby generating inpainting results that are authentic and credible in both structural and semantic aspects. For instance, Fig. 5(g) demonstrates that ours has accurately inpainted the ascending shark; in image inpainting tasks with strong structural features (e.g., Figs. 5(a) and 5(c)), by virtue of the structural guidance mechanism, ours has also achieved good restoration of image structures.

To explore the critical inpainting capability of the model under high porosity, the Inp-JSRDiff method was used to inpaint the PSV test dataset under porosities of $[0.6, 0.7)$, $[0.7, 0.8)$, and $[0.8, 0.9)$. Table 3 provides a quantitative analysis of the inpainting performance under different values of T and different high-porosity levels. As porosity increases to the extreme interval $[0.8, 0.9)$, FID rises sharply for all T , indicating clear perceptual degradation due to the severely under-constrained inpainting when only limited context is observed. Accordingly, we consider porosity < 0.8 as the practical operating range where the method remains more reliable.

Table 3: COMPARISON OF SAMPLING STEPS T FOR EXTREME HIGH-POROSITY INPAINTING ON PSV DATASET.

Mask ratio	[0.6,0.7)		[0.7,0.8)		[0.8,0.9)	
	PSNR $\uparrow$	FID $\downarrow$	PSNR $\uparrow$	FID $\downarrow$	PSNR $\uparrow$	FID $\downarrow$
$T = 1$	20.73	100.61	19.53	121.68	17.94	186.76
$T = 2$	20.79	108.81	19.69	125.67	17.98	192.76
$T = 3$	20.86	118.35	20.06	138.95	17.99	200.97

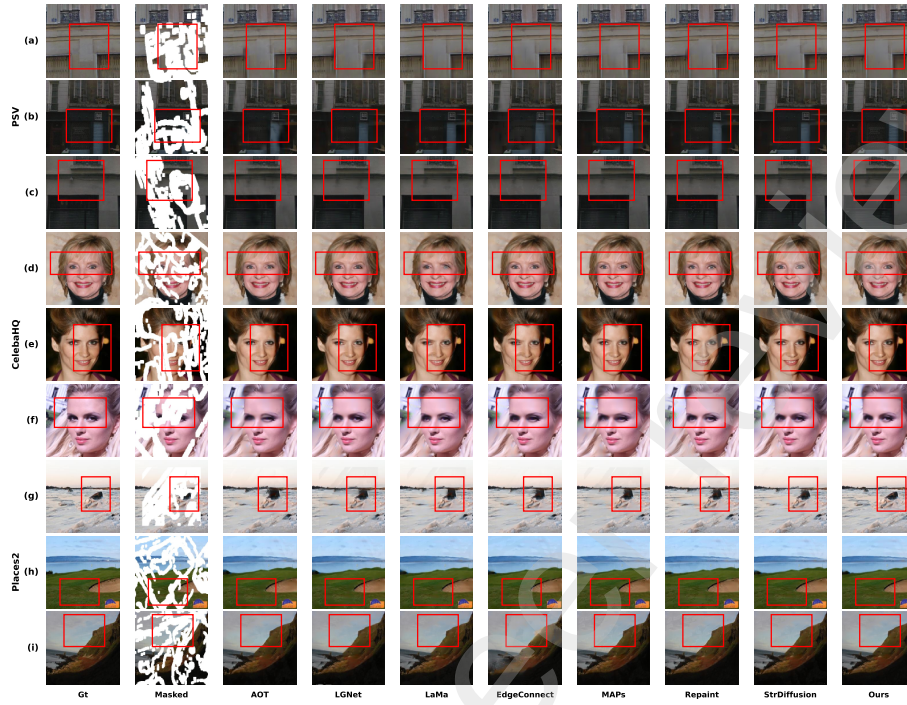


Fig. 5: Visual comparison of inpainting results across different methods on PSV, CelebA-HQ, and Places2 datasets. (a)–(h) Different inpainting scenarios ranging from architectural structures to natural scenes. Our proposed method (rightmost column) demonstrates superior structural consistency, detail preservation, and semantic coherence compared to existing approaches. Note how our method accurately reconstructs fine details like the shark head in (g) and preserves natural color transitions in vegetation areas (h) while avoiding artifacts present in other methods.

4.3. Ablation Study

To examine whether the reversible (invertible) design brings quality degradation or training instability, we perform a controlled ablation by enabling/disabling the reversible component while keeping other factors unchanged. Experiments are conducted on the PSV dataset using only the reversible diffusion backbone with a pure L_1 loss. We set batch size to 4, gradient accumulation to 1, and learning rate to 1×10^{-4} , and evaluate two step settings $T \in \{1, 2\}$ on a single 48GB GPU. As shown in Table 4, the reversible component significantly reduces peak training memory (e.g., 19.4 GB $\rightarrow$ 11.4 GB for $T = 1$, 37.5 GB $\rightarrow$ 11.5 GB for $T = 2$), while maintaining comparable inpainting performance under both low-porosity [0, 0.4) and high-porosity [0.4, 0.6) conditions (PSNR/FID differences are marginal). Training curves in Fig. 6

further indicate stable convergence for both $T = 1$ and $T = 2$, suggesting that reversibility does not compromise optimization stability. The main trade-off is increased training time due to recomputation (e.g., 8 h $\rightarrow$ 12 h for $T = 1$, 8 h $\rightarrow$ 26 h for $T = 2$), confirming a clear memory-compute exchange.

Table 4: REVERSIBLE VS. NON-REVERSIBLE ABLATION ON PSV: MEMORY USAGE, TRAINING TIME, AND PSNR/FID UNDER POROSITY RANGES FOR $T = 1, 2$. $\uparrow$ HIGHER IS BETTER. $\downarrow$ LOWER IS BETTER.

Component	T	Memory (GB)	Train time (h)	[0.1,0.4]		[0.4,0.6]	
				PSNR $\uparrow$	FID $\downarrow$	PSNR $\uparrow$	FID $\downarrow$
No reversible	1	37.5	8	28.79	81.0	22.74	70.88
	2	19.4	8	29.31	28.82	23.07	73.16
Reversible	1	11.4	12	28.86	28.14	22.98	70.96
	2	11.5	26	29.24	28.97	23.02	73.12

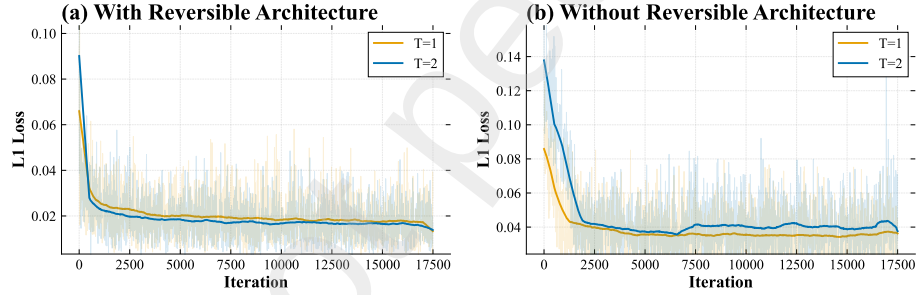


Fig. 6: L1 training loss with/without the reversible architecture on PSV for $T = 1, 2$.

We verify the effectiveness of CADI and analyze its scene-adaptive weighting behavior. To isolate the contribution of attention-based dynamic injection, we fix all other hyper-parameters and set $T = 1$ on the PSV dataset. We use the edge structures restored by E-StructRM as guidance conditions and keep the same injector placement, while varying only the attention mechanism: (i) no attention baseline, (ii) channel attention (CADI, ours), (iii) spatial attention, and (iv) channel+spatial attention. We report the additional attention parameters and FLOPs, and evaluate inpainting quality under low-porosity [0.1, 0.4] and high-porosity [0.4, 0.6] masks using PSNR/FID. As shown in Table 5, channel attention achieves a clear gain over the no-attention baseline with negligible overhead: PSNR increases from 27.86 $\rightarrow$ 29.17 (+1.31 dB, $\approx$ 4.7%) at

low porosity and $22.00 \rightarrow 23.17$ (+1.17 dB, $\approx 5.3\%$) at high porosity, while FID drops from $29.64 \rightarrow 27.89$ and $73.85 \rightarrow 69.82$, respectively. This improvement is obtained with only 0.152 K attention parameters and a small FLOPs increase ($6.55 \text{ M} \rightarrow 7.08 \text{ M}$, $\approx 8.1\%$). In contrast, spatial attention yields slightly worse quality but incurs a much higher compute cost (13.89 M FLOPs, +112% vs. baseline). Channel+spatial attention brings only marginal further improvement but substantially increases parameters and FLOPs, making it less cost-effective.

Table 5: ABLATION STUDY ON DIFFERENT ATTENTION MECHANISMS. COMPARING THE EFFECT OF DIFFERENT ATTENTION STRATEGIES ON INPAINTING QUALITY AND COMPUTATIONAL OVERHEAD. $\uparrow$ HIGHER IS BETTER. $\downarrow$ LOWER IS BETTER.

Attention	Parameters (K)	FLOPs (M)	Low Porosity [0.1,0.4]		High Porosity [0.4,0.6]	
			PSNR $\uparrow$	FID $\downarrow$	PSNR $\uparrow$	FID $\downarrow$
Baseline	0.00	6.55	27.86	29.64	22.00	73.85
Channel attention	0.152	7.08	29.17	27.89	23.17	69.28
Spatial	0.200	13.89	28.06	27.93	22.84	70.69
SpatioTemporal	0.354	13.89	29.19	27.78	23.35	68.96

In addition, to verify the scene-adaptive weighting behavior of CADI, we visualize its channel-wise weights at inference time using four images randomly sampled from the Paris StreetView (PSV) test set, as shown in Fig. 7. We inspect four representative blocks spanning both the encoder and decoder, i.e., DB1-1, DB2-1, UB2-0, and UB1-1 (cf. Fig. 3), to more comprehensively examine the stage-wise weighting dynamics. Overall, the learned weights exhibit a clear stage-wise evolution: in the downsampling blocks (DB), the four sources are weighted in a relatively balanced manner, reflecting generic feature extraction before structure-texture specialization. When entering the early upsampling block (UB2-0), the edge-prior channel consistently becomes more prominent ($\text{Ch4} \approx 0.55$), indicating that CADI tends to emphasize structural guidance when recovering geometry and boundaries. In the later refinement block (UB1-1), the fusion shifts toward mask-related cues, and the dominant channel varies with scene content. For example, at UB1-1, Image 1 (a largely planar wall with weak structural boundaries) assigns a higher weight to masked-input features ($\text{Ch3} = 0.63$), whereas Image 2 (foliage with richer fine textures) shows a more distributed pattern with reduced reliance on Ch3 ($\text{Ch3} = 0.51$). Similarly, Image 4 (bridge rails/horizon lines

with a strong global layout) yields an even stronger emphasis on Ch3 (Ch3 = 0.64). These observations indicate that CADI does not use fixed fusion coefficients; instead, it learns dynamic, content-dependent channel re-weighting across network stages and scene conditions.

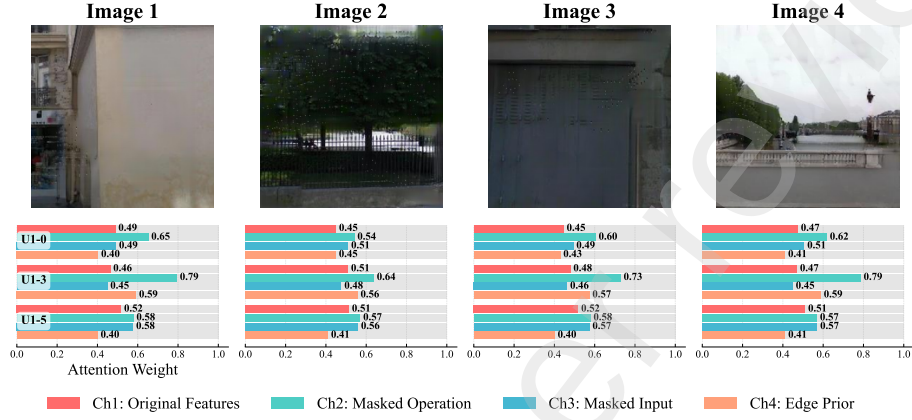


Fig. 7: Visualization of CADI’s channel-wise gating weights on four randomly sampled PSV images. For each image, we report the weights at four representative blocks (DB1-1, DB2-1, UB2-0, and UB1-c; cf. Fig. 3), spanning both encoder and decoder stages. CADI outputs a 4-D weight vector $\mathbf{W} = [w_1, w_2, w_3, w_4] \in (0, 1)^4$ to modulate four injected sources: (Ch1) original backbone features, (Ch2) masked operation features, (Ch3) masked-input features, and (Ch4) structural guidance (edge priors). The cross-image and cross-block variations illustrate that CADI performs stage-dependent and content-adaptive fusion rather than using fixed coefficients.

We also verify the effectiveness of the sparse spatial structure in guiding image inpainting. We inject four types of edge structures into the U-Net network of the RevDiffInpaint via CADI for structural guidance during the inpainting process, including edge structures of complete images, edge structures restored by the E-StructRM, edge structures of occluded images, and empty structures. Table 6 indicates the effects of different structural guidance settings on inpainting results, demonstrating that our method—whereby we restore the edges of occluded images via the E-StructRM and then inject these restored edges as guidance conditions into the RevDiffInpaint—is effective.

Then, we further analyze whether end-to-end optimization improves the coupling between structure restoration and inpainting. Under $T = 1$, we use a pre-trained E-

Table 6: ABLATION STUDY ON DIFFERENT EDGE PRIOR SETTINGS. COMPARING THE EFFECT OF DIFFERENT EDGE PRIORS ON IMAGE INPAINTING QUALITY METRICS. $\uparrow$ HIGHER IS BETTER. $\downarrow$ LOWER IS BETTER.

Metrics	Mask ratio	Edge prior type			
		No prior	Masked	Inpainted	Original
PSNR $\uparrow$	[0.1,0.4)	27.07	28.20	29.17	30.56
	[0.4,0.6)	21.37	22.40	23.17	24.11
FID $\downarrow$	[0.1,0.4)	39.40	35.90	27.89	23.24
	[0.4,0.6)	79.37	74.90	69.28	59.41

StructRM and compare two settings: Freezing, where E-StructRM is fixed during inpainting training, and Activation, where E-StructRM is jointly optimized end-to-end. In addition, we vary the injection location of structure guidance into the diffusion backbone: DB1/UB1 (shallow), DB2/UB2 (deep), and ALL (all blocks). As reported in Table 7, enabling end-to-end optimization (Activation) consistently improves restoration quality over freezing E-StructRM across both porosity regimes, with more pronounced gains under high-porosity masks (e.g., PSNR (23.17) vs. (22.81); SSIM (0.768) vs. (0.747); LPIPS (0.190) vs. (0.203); and FID (69.28) vs. (71.24)). Moreover, injecting structural guidance into all diffusion blocks (ALL) yields the best overall and most robust performance (e.g., the highest PSNR/SSIM and lowest FID in both regimes), indicating that effective structure-texture synergy benefits from multi-scale coupling throughout the diffusion hierarchy rather than a single injection point.

Table 7: END-TO-END COUPLING ABLATION ON PSV ($T=1$): FREEZING VS. ACTIVATING E-STRUCTRM AND INJECTION LOCATION. $\uparrow$ HIGHER IS BETTER. $\downarrow$ LOWER IS BETTER.

E-StructRM Weight Status	Inject Location	Low Porosity [0.1,0.4)				High Porosity [0.4,0.6)			
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
Freezing	DB1/UB1	28.94	0.924	0.069	28.34	22.68	0.744	0.205	71.24
	DB2/UB2	28.75	0.926	0.071	28.27	22.59	0.747	0.203	71.67
	ALL	29.07	0.926	0.070	28.04	22.81	0.746	0.204	71.32
Activation	DB1/UB1	29.08	0.927	0.067	28.13	22.75	0.754	0.198	70.54
	DB2/UB2	28.95	0.924	0.069	27.94	22.86	0.761	0.187	69.87
	ALL	29.17	0.931	0.068	27.89	23.17	0.768	0.190	69.28

Finally, we verify the inhibitory effect of introducing a discriminator to RevDiffInpaint on excessive smoothing that is caused by changes in optimization objectives. By comparing the quantitative inpainting results of models trained without a discriminator and models trained with a discriminator, as shown in Table 8, it can be seen that

the model without a discriminator has a lower L_1 but higher perceptual metrics (LPIPS, FID), and fails to exhibit better perceptual performance.

Table 8: EFFECT OF DISCRIMINATOR. COMPARING PERFORMANCE METRICS ACROSS DIFFERENT MASK RATIOS. $\uparrow$ HIGHER IS BETTER. $\downarrow$ LOWER IS BETTER.

Discriminator	Metrics	Mask ratio				
		[0.1,0.2)	[0.2,0.3)	[0.3,0.4)	[0.4,0.5)	[0.5,0.6)
No	PSNR $\uparrow$	32.51	28.87	26.95	24.87	22.07
	L1(10^{-2}) $\downarrow$	1.03	1.62	2.58	3.37	5.08
	FID $\downarrow$	16.31	29.58	48.63	64.82	90.06
Yes	PSNR $\uparrow$	32.24	29.01	26.27	24.43	21.90
	L1(10^{-2}) $\downarrow$	1.05	1.82	2.73	3.79	5.71
	FID $\downarrow$	13.96	26.45	43.27	57.97	80.58

5. Discussion

5.1. Impact of T Value on Model Mechanism

To investigate the impact of sampling step T on the model mechanism, we analyze the reversible diffusion architecture in RevDiffInpaint under the following settings: a resolution of 256, batch size of 1, FP32 precision, and without gradient checkpointing. This section first analyzes the variation patterns of model parameter count, peak training memory, per-step FLOPs, and inference time per image under different T values from the perspective of model scale, computational efficiency, and resource consumption. Then, from the frequency domain perspective, it examines the regulatory effect of T on the spectral features of inpainted images, so as to further explain the root cause of the model’s performance differences in pixel-level fidelity and perceptual quality. As shown in Table 9, our model is highly parameter-efficient: the total number of parameters remains almost unchanged as the sampling steps T increase, rising only from 115.26 M at $T = 1$ to 115.34 M at $T = 10$ (+0.08 M, $\approx 0.07\%$). This is because the overwhelming majority of parameters come from the shared U-Net backbone, while the only T -dependent parts are the step-specific injectors and the channel-attention modules. Although these lightweight modules grow linearly with T across 20 injector locations, the per-step overhead is only about 9 K parameters (mainly 1×1 convolutions and a small MLP), making their contribution to the overall parameter count negligible. Consequently, adjusting T provides a flexible quality-efficiency trade-off without introducing a proportional increase in parameters or memory footprint.

T directly affects the model’s computational load and hardware requirements. To quantify its impact, we tested key performance indicators under different T values on the NVIDIA RTX 4070 Ti Super GPU. As shown in Table 10, the reversible structure design (Section 3.1.3) reduces the GPU memory cost for intermediate activation reconstruction to a constant level, so the peak training GPU memory hardly increases with T . Per-step FLOPs is fixed (0.23 G), and the total computation is $0.23 \times T$ (GFLOPs). Since the noise prediction network needs to be fully executed at each step, macroscopically, the inference time increases linearly with T : $t_{\text{infer}} \approx 329.6 \times T$ (ms). However, microscopically, the time consumption of each component in forward propagation (inference process) is not proportional to T . Table XI decomposes the components used during inference into DiffusionStep, inverse components, and CADI, and quantitatively analyzes the inference time consumption of each component under different T values. As T increases, the proportion of time consumption of DiffusionStep gradually increases, the model’s influence by first-order reversibility becomes increasingly significant, while the influence of second-order reversibility gradually diminishes.

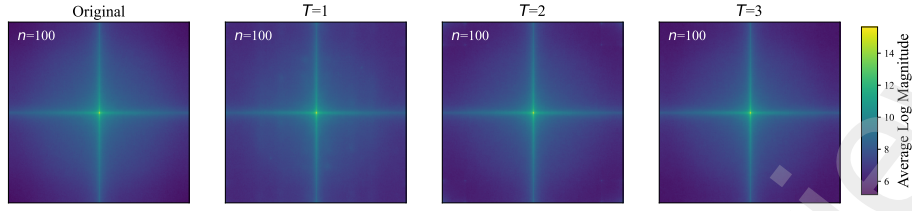
Table 9: COMPUTATIONAL EFFICIENCY METRICS UNDER DIFFERENT SAMPLING STEPS T . MEASURING THE TRADE-OFF BETWEEN SAMPLING STEPS AND COMPUTATIONAL COST.

T Value	Parameters (M)	Memory Usage (MB)	Peak Memory (MB)	FLOPs (G)	Infer-Time (ms)
1	115.26	2401.40	2976.58	0.23	329.6
2	115.27	2412.77	3494.37	0.46	658.2
4	115.28	2408.74	3492.37	0.92	1319.8
6	115.30	2410.78	3493.41	1.38	1984.2
8	115.32	2410.39	3493.54	1.85	2652.0
10	115.34	2410.20	3494.99	2.31	3336.4

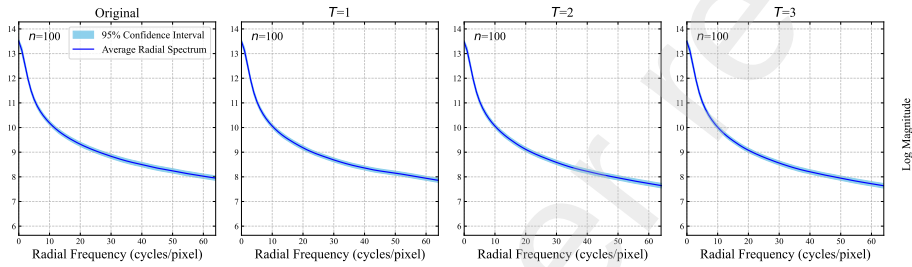
Table 10: DECOMPOSITION OF INFERENCE TIME FOR DIFFERENT COMPONENTS UNDER VARYING SAMPLING STEPS T . SHOWING THE TIME COST AND PROPORTION OF EACH COMPONENT.

T Value	Diffusion Step		Reversible component		Channel attention	
	Time (ms)	Prop (%)	Time (ms)	Prop (%)	Time (ms)	Prop (%)
1	170.04	51.0	161.85	48.5	1.71	0.5
2	449.68	67.5	214.02	32.1	2.05	0.3
3	760.12	75.7	241.18	24.0	2.22	0.2

We analyze the frequency domain properties of the RevDiffInpaint model on 100 test images from the PSV dataset (with mask porosity fixed in the range $[0.5, 0.6]$), showing that fewer sampling steps $T = 1$ optimize perceptual metrics, while more



(a) Average logarithmic magnitude spectrum heatmap



(b) Radial average spectrum curve

Fig. 8: Frequency domain feature analysis of inpainting results on the PSV test set ($n = 100$) under the same T value.

sampling steps $T > 1$ optimize pixel-level metrics. Figure 8 shows that at $T = 1$, the model has stronger energy in the high-frequency regions (bright yellow), indicating better detail preservation. At $T = 3$, more low-frequency energy is concentrated at the spectrum center, suggesting better structural consistency. Figure 8(b) shows the radial average spectrum curve, where the curve for $T = 1$ maintains a steeper slope in the high-frequency region compared to $T = 3$, emphasizing that fewer steps preserve more high-frequency details, while more steps focus on low-frequency structures. This frequency domain analysis confirms that $T = 1$ optimizes perceptual metrics (LPIPS/FID) by preserving high-frequency detail energy, while $T > 1$ optimizes pixel-level metrics (SSIM/PSNR) by prioritizing low-frequency structural error convergence.

5.2. Model Inadequacy

For image inpainting, the proposed method achieves better results via single or very few sampling steps and has lower overall time. However, single-step sampling of this method in inference takes longer (330 ms on average), significantly higher than

Repaint (78 ms) and StrDiffusion (68 ms). Two main reasons account for this. First, the U-Net in the SDv1.5 model (even after pruning) has more complex attention mechanisms and residual connections than that in the DDPM adopted by Repaint. Second, the model introduces additional components for reversibility. Although training does not require storing the entire computational graph and thus reduces hardware memory requirements, recalculating intermediate results during inference significantly increases single-step inference time. Thus, the reversible structure is a time-for-space trade-off scheme that incurs significant time cost. Future work will focus on reducing memory requirements during training while improving single-step inference speed to achieve synergistic optimization of training and inference.

6. Conclusion

Addressing the core issues of poor global consistency, blurry details, and high training and inference costs in existing image inpainting methods under large-area missing scenarios, this paper proposes a novel inpainting scheme, namely Inp-JSRDiff. Different from recent structure-guided diffusion inpainting approaches that primarily enhance denoising via structural guidance under multi-step sampling, our approach aims to achieve few-step, end-to-end trainable inpainting with strict known-pixel consistency and practical memory efficiency. First, an end-to-end reversible diffusion framework is introduced into inpainting tasks. We reinterpret deterministic sampling as an unrolled reconstruction network with hard known-pixel clamping and learnable step coefficients $\{\alpha_t\}$, shifting the optimization objective from noise prediction to direct masked image reconstruction; the insertion of reversible modules further alleviates the strong hardware dependency of conventional diffusion training. Second, CADI is designed to achieve multi-level dynamic fusion of texture and edge priors within the diffusion network, enhancing the structural integrity and detail fidelity of the restored content. Third, by jointly training the edge restoration network and the diffusion inpainting network under a unified objective, we enable differentiable cross-module gradient propagation and strengthen global texture-edge semantic consistency. Although the current single-step sampling time remains higher than that of some inpainting approaches, future work will focus on accelerating inference by lightweighting the noise-prediction

backbone and reducing redundant computations (e.g., efficient architecture design and efficient loss evaluation), while maintaining the few-step reconstruction quality.

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References

- [1] A. Criminisi, P. Perez, K. Toyama, *Object removal by exemplar-based inpainting*, in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2003, pp. 1–8.
- [2] J. Liang, H. Zeng, M. Cui, X. Xie, L. Zhang, *PPR10K: A large-scale portrait photo retouching dataset with human-region mask and group-level consistency*, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 653–661.
- [3] F. Lu, Z. Zhang, T. Liu, C. Tang, H. Bai, G. Zhai, J. Chen, X. Wu, *A weakly supervised inpainting-based learning method for lung CT image segmentation*, Pattern Recognit., vol. 144, 2023, Art. no. 109861.
- [4] G. Gallo, F. Stanco, S. Battaino, *Digital Imaging for Cultural Heritage Preservation: Analysis, Restoration and Reconstruction of Ancient Artworks*, Taylor and Francis, 2011.
- [5] Y. Yang, X. Chen, J. Liao, *Uni-paint: A Unified Framework for Multi-modal Image Inpainting with Pretrained Diffusion Model*, in Proceedings of the 31st ACM International Conference on Multimedia (MM '23), Ottawa, ON, Canada, 2023, pp. 3190–3199, doi: 10.1145/3581478.3612200.

- [6] M. Bertalmio, G. Sapiro, V. Caselles, C. Ballester, *Image inpainting*, in Proc. ACM SIGGRAPH, 2000, pp. 417–424.
- [7] C. Ballester, M. Bertalmio, V. Caselles, G. Sapiro, J. Verdera, *Filling-in by joint interpolation of vector fields and gray levels*, IEEE Trans. Image Process, vol. 10, no. 8, pp. 1200–1211, 2001.
- [8] C. Barnes, E. Shechtman, A. Finkelstein, D. B. Goldman, *PatchMatch: A randomized correspondence algorithm for structural image editing*, ACM Trans. Graph., vol. 28, no. 3, p. 24, 2009.
- [9] S. Darabi, E. Shechtman, C. Barnes, D. B. Goldman, P. Sen, *Image melding: combining inconsistent images using patch-based synthesis*, ACM Trans. Graph. (Proc. SIGGRAPH), vol. 31, no. 4, pp. 1–10, 2012.
- [10] H. Xiang, Q. Zou, M. A. Nawaz, X. Huang, F. Zhang, H. Yu, *Deep learning for image inpainting: A survey*, Pattern Recognition, vol. 134, 2023, Art. no. 109046, doi: 10.1016/j.patcog.2022.109046.
- [11] G. Liu, F. A. Reda, K. J. Shih, T.-C. Wang, A. Tao, B. Catanzaro, *Image inpainting for irregular holes using partial convolutions*, in Proc. Eur. Conf. Comput. Vision (ECCV), Sep. 2018.
- [12] J. Yu, Z. Lin, J. Yang, et al., *Free-form image inpainting with gated convolution*, in Proc. IEEE Int. Conf. Comput. Vision, 2019, pp. 4471–4480.
- [13] L. Chi, B. Jiang, Y. Mu, *Fast Fourier convolution*, in Adv. Neural Inf. Process. Syst., vol. 33, pp. 4479–4488, 2020.
- [14] K. Nazeri, E. Ng, T. Joseph, F. Qureshi, M. Ebrahimi, *EdgeConnect: Structure Guided Image Inpainting using Edge Prediction*, in Proc. IEEE/CVF Int. Conf. Comput. Vision (ICCV) Workshops, Oct. 2019.
- [15] C. Liu, S. Xu, J. Peng, K. Zhang, D. Liu, *Toward Interactive Image Inpainting via Robust Sketch Refinement*, IEEE Trans. Multimedia, vol. 26, pp. 9973–9987, 2024, doi: 10.1109/TMM.2024.3402620.

- [16] Y. Wang, D. Guo, H. Zhao, M. Yang, H. Zheng, *Image inpainting via multi-scale adaptive priors*, Pattern Recognit., vol. 162, pp. 111410, Jun. 2025, doi: 10.1016/j.patcog.2025.111410.
- [17] J. Ho, A. Jain, P. Abbeel, *Denoising diffusion probabilistic models*, in Proc. Adv. Neural Inf. Process. Syst., vol. 33, pp. 6840–6851, 2020.
- [18] A. Lugmayr, M. Danelljan, A. Romero, et al., *Repaint: Inpainting using denoising diffusion probabilistic models*, in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2022, pp. 11461–11471.
- [19] H. Liu, Y. Wang, B. Qian, M. Wang, Y. Rui, *Structure Matters: Tackling the Semantic Discrepancy in Diffusion Models for Image Inpainting*, in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), Jun. 2024, pp. 8038–8047.
- [20] C. Corneau, R. Gadde, A. M. Martinez, *LatentPaint: Image Inpainting in Latent Space with Diffusion Models*, in Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV), 2024, pp. 4322–4331, doi: 10.1109/WACV57701.2024.00428.
- [21] B. Chen et al., *Invertible Diffusion Models for Compressed Sensing*, IEEE Trans. Pattern Anal. Mach. Intell., vol. 47, no. 5, pp. 3992–4006, May 2025, doi: 10.1109/TPAMI.2025.3538896.
- [22] T. Karras, T. Aila, S. Laine, J. Lehtinen, *Progressive growing of GANs for improved quality, stability, and variation*, in Int. Conf. Learn. Represent., 2018.
- [23] B. Zhou, A. Lapedriza, A. Khosla, A. Oliva, A. Torralba, *Places: A 10 million image database for scene recognition*, IEEE Trans. Pattern Anal. Mach. Intell., vol. 40, no. 6, pp. 1452–1464, 2017.
- [24] C. Doersch, S. Singh, A. Gupta, J. Sivic, A. A. Efros, *What makes paris look like paris?* ACM Trans. Graph., vol. 31, no. 4, pp. 101:1–101:9, 2012.
- [25] Q. Liu, Y. Jiang, Z. Tan, D. Chen, Y. Fu, Q. Chu, G. Hua, N. Yu, *Transformer-based pluralistic image completion with reduced information loss*, IEEE Trans. Pattern Anal. Mach. Intell., 2024.

- [26] J. Yu, Z. Lin, J. Yang, X. Shen, A. Lu, T. S. Huang, *Generative image inpainting with contextual attention*, in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 5505–5514.
- [27] R. Rombach, A. Blattmann, D. Lorenz, et al., *High-resolution image synthesis with latent diffusion models*, in Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022: 10684-10695.
- [28] X. Guo, H. Yang, D. Huang, *Image Inpainting via Conditional Texture and Structure Dual Generation*, in Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), Oct. 2021, pp. 14134–14143.
- [29] Q. Dong, C. Cao, Y. Fu, *Incremental Transformer Structure Enhanced Image Inpainting with Masking Positional Encoding*, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022.
- [30] Y. Zeng, J. Fu, H. Chao, B. Guo, *Aggregated contextual transformations for high-resolution image inpainting*, IEEE Trans. Vis. Comput. Graphics, vol. 29, no. 7, pp. 3266–3280, 2022.
- [31] S. Iizuka, E. Simo-Serra, H. Ishikawa, *Globally and locally consistent image completion*, ACM Trans. Graph. (Proc. SIGGRAPH), vol. 36, no. 4, pp. 1–14, 2017.
- [32] W. Quan, R. Zhang, Y. Zhang, Z. Li, J. Wang, D.-M. Yan, *Image inpainting with local and global refinement*, IEEE Trans. Image Process., vol. 31, 2022, pp. 2405–2420.
- [33] J. Song, C. Meng, S. Ermon, *Denoising Diffusion Implicit Models*, in Proceedings of International Conference on Learning Representations (ICLR), 2021.
- [34] Y. Wang, J. Yu, J. Zhang, *Zero-Shot Image Restoration Using Denoising Diffusion Null-Space Model*, in Proceedings of International Conference on Learning Representations (ICLR), 2023.
- [35] T.-C. Wang, M.-Y. Lin, J.-Y. Zhu, A. Tao, J. Kautz, B. Catanzaro, *High-resolution image synthesis and semantic manipulation with conditional GANs*, in Pro-

- ceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018, p. 5.
- [36] R. Suvorov, E. Logacheva, A. Mashikhin, A. Remizova, A. Ashukha, A. Silvestrov, N. Kong, H. Goka, K. Park, V. Lempitsky, *Resolution-robust large mask inpainting with Fourier convolutions*, in Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2022, pp. 2149–2159.
 - [37] Z. Wang, A. C. Bovik, H. R. Sheikh, E. P. Simoncelli, *Image quality assessment: from error visibility to structural similarity*, IEEE Trans. Image Process., vol. 13, no. 4, pp. 600–612, 2004.
 - [38] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, S. Hochreiter, *GANs trained by a two-time-scale update rule converge to a local Nash equilibrium*, in Advances in Neural Information Processing Systems, 2017, pp. 6626–6637.
 - [39] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, O. Wang, *The unreasonable effectiveness of deep features as a perceptual metric*, in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 586–595.
 - [40] N. Li, L. Wang, B. Zhang, D. Tao, *Recurrent feature reasoning for image inpainting*, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 7760–7769.
 - [41] Q. Zhang, X. Liu, H. Zhang, *Nonlocal low-rank tensor completion for visual data*, IEEE Transactions on Cybernetics, vol. 50, no. 7, pp. 3268–3281, 2020.